THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. According to the particulate model, matter is made up of: 1

(a) continuous jelly-like material **(b)** tiny particles
(c) only air **(d)** only liquids

2. The change from solid to liquid is called: 1

(a) condensation **(b)** freezing **(c)** melting **(d)** boiling

3. Which of the following increases the rate of evaporation? 1

(a) absence of wind **(b)** decrease in surface area
(c) increase in humidity **(d)** increase in temperature

4. There are spaces between the particles of matter. Which observation supports this? 1

(a) wood burns
(b) ice floats
(c) metals shine
(d) sugar dissolves in water without raising the water level much

5. The change from liquid to gas at any temperature below the boiling point is: 1

(a) sublimation **(b)** melting **(c)** evaporation **(d)** freezing

6. Wet clothes dry faster when: 1

(a) kept in damp room **(b)** tied tightly
(c) folded and kept in shade **(d)** spread out in a windy place

7. The particles of matter are: 1

(a) always at rest in solids **(b)** constantly moving
(c) completely still **(d)** stuck together permanently

8. The temperature at which a solid melts at normal pressure is its: 1

(a) room temperature **(b)** boiling point
(c) melting point **(d)** freezing point of gases

9. Evaporation causes: 1

(a) heating **(b)** burning **(c)** cooling **(d)** no change

10. The spreading of a drop of ink in water on its own is called: 1

(a) melting **(b)** sublimation **(c)** diffusion **(d)** evaporation

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. State two assumptions of the particulate model of matter. 2

12. State two differences between evaporation and boiling. 2

13. Name the process in the formation of dew. 2

14. What is diffusion? Give one example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show that particles of matter have spaces between them. 3

16. Explain the cooling effect of evaporation using particles. 3

17. Why do we see water droplets on the outside of a glass of cold water? 3

OR

Describe an activity to show diffusion in liquids. 3

18. Describe how a gas can be changed to a liquid. 3

OR

How is a heating curve of water explained? Describe the flat portions. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the particulate nature of matter with the help of three activities. 5

OR

Describe an experiment to study the cooling by evaporation using cotton wool dipped in different liquids on a thermometer bulb. 5

20. Differentiate between solids, liquids and gases in five points. 5

OR

Explain why a gas exerts pressure, using the particulate model. 5

END OF QUESTION PAPER — Total: 40 Marks

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